THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Power Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 2^8	$2 \times 2^7 = 2^8$
2	**(a)** 4	$2x + 1 = 9$
3	**(a)** 7	$2^4 \times 5^7 = 10^4 \times 125$
4	**(a)** 3	Ones digits repeat in the cycle [7, 9, 3, 1]
5	**(a)** 27	$x = 5$, so 3^3
6	**(c)** $\frac{6}{5}$	$\left(\frac{1}{2} + \frac{1}{3}\right)^{-1}$
7	**(d)** 0.5×10^6	Convert each to usual form and compare
8	**(d)** 40	$2^{x+3} = 2^x \times 2^3$
9	**(a)** Both A and R are true and R is the correct explanation of A.	$2^{-3} = \frac{1}{2^3} = \frac{1}{8}$; R is exactly the rule used.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	An odd power of -1 is -1, so A is true; R is also true but is unrelated to the value of $(-1)^{101}$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$243 = 3^5$ (1); $2x - 1 = 5$ gives $x = 3$ (1).	1+1
12	Convert to the same power or usual form: 340000 (1); 3.4×10^5 (1).	1+1
13	Ones digits of powers of 2 repeat as [2, 4, 8, 6] (1); 47 leaves position 3 in the cycle, so the digit is 8 (1).	1+1
14	$16384 = 2^{14}$ (1); $3x + 2 = 14$ gives $x = 4$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Thickness doubles at each fold: 0.1×2^{10} (1); $= 102.4$ mm (1); in cm $= 10.24$ cm (1).	1+1+1
16	2 min $= 120$ s (1); distance $= 3 \times 10^8 \times 120$ (1); $=3.6 \times 10^{10}$ m (1).	1+1+1
17	Take 2^5 common in the numerator: $2^5(2 + 1)$ (1); cancel $(2 + 1)$ (1); $2^5 \div 2^4 = 2$ (1).	1+1+1
	OR: Thickness doubles at each fold: 0.2×2^8 (1); $= 51.2$ mm (1); in cm $= 5.12$ cm (1).	1+1+1
18	5 min $= 300$ s (1); distance $= 3 \times 10^8 \times 300$ (1); $=9 \times 10^{10}$ m (1).	1+1+1
	OR: Take 3^4 common in the numerator: $3^4(3 + 1)$ (1); cancel $(3 + 1)$ (1); $3^4 \div 3^3 = 3$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 1.5×10^8 km (1). (ii) Time $= \frac{1.5 \times 10^8}{3 \times 10^5} = 0.5 \times 10^3 = 500$ s (1). (iii) 5×10^2 s (1). (iv) Distance $= 3 \times 10^5 \times 1.3 = 3.9 \times 10^5$ km (1+1).	1+1+1+2
20	Number $= 2 \times 4^3$ (2); value 128 (2); it is $4^3 = 64$ times the original (1).	2+2+1
	OR: A is heavier (1); ratio $= \frac{9}{3} \times 10^{12-9} = 3000$ times (2); total 9.003×10^{12} kg (2).	1+2+2

Mistakes to avoid:

- Writing $2^{-3} = -8$ or $-\frac{1}{8}$; a negative exponent means the **reciprocal**, so $2^{-3} = \frac{1}{8}$.
- Adding the bases: $2^3 \times 2^4 = 4^7$ instead of $2^{3+4} = 2^7$.
- Saying $a^0 = 0$; for $a \neq 0$ it is 1.
- Multiplying exponents when multiplying powers of the same base ($a^m \times a^n = a^{mn}$) or adding them in $(a^m)^n$.
- Leaving the answer as 25×10^4 or 0.25×10^6; in standard form the number part must satisfy $1 \leq m < 10$.
- Getting the sign of the exponent wrong for small numbers: $0.0003 = 3 \times 10^{-4}$, not 3×10^4.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Power Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 9	$2^6 \times 5^9 = 10^6 \times 125$
2	**(d)** 9	Ones digits repeat in the cycle [3, 9, 7, 1]
3	**(a)** 32	$x = 9$, so 2^5
4	**(a)** 2	$\left(\frac{1}{3} + \frac{1}{6}\right)^{-1}$
5	**(b)** 0.9×10^{-2}	Convert each to usual form and compare
6	**(c)** 36	$3^{x+2} = 3^x \times 3^2$
7	**(d)** 4^7	$4 \times 4^6 = 4^7$
8	**(b)** 5	$2x + 3 = 13$
9	**(c)** A is true but R is false.	$3^0 + 2^0 = 1 + 1 = 2$ so A is true; R is false because $a^0 = 1$.
10	**(d)** A is false but R is true.	$2^3 \times 3^3 = 6^3 = 216$, not $5^3 = 125$; A is false, R is a true law.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Convert to the same power or usual form: 3000000 (1); 3×10^6 (1).	1+1
12	Ones digits of powers of 4 repeat as [4, 6] (1); 61 leaves position 1 in the cycle, so the digit is 4 (1).	1+1
13	$125 = 5^3$ (1); $2x - 3 = 3$ gives $x = 3$ (1).	1+1
14	Convert to the same power or usual form: 0.073 (1); 7.3×10^{-2} (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Thickness doubles at each fold: 0.05×2^{12} (1); $= 204.8$ mm (1); in cm $= 20.48$ cm (1).	1+1+1
16	8 min $= 480$ s (1); distance $= 3 \times 10^8 \times 480$ (1); =1.44×10^{11} m (1).	1+1+1
17	Take 5^3 common in the numerator: $5^3(5 + 1)$ (1); cancel $(5 + 1)$ (1); $5^3 \div 5^2 = 5$ (1).	1+1+1
	OR: Thickness doubles at each fold: 0.4×2^6 (1); $= 25.6$ mm (1); in cm $= 2.56$ cm (1).	1+1+1
18	12 min $= 720$ s (1); distance $= 3 \times 10^8 \times 720$ (1); =2.16×10^{11} m (1).	1+1+1
	OR: Take 4^6 common in the numerator: $4^6(4 + 1)$ (1); cancel $(4 + 1)$ (1); $4^6 \div 4^5 = 4$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $5 \times 2 = 10$ (1). (ii) $5 \times 2^6 = 5 \times 64 = 320$ (1). (iii) After 7 h: $5 \times 128 = 640$; after 8 h: $5 \times 256 = 1280 > 1000$, so 8 hours (1). (iv) $\frac{5 \times 2^8}{5 \times 2^4} = 2^{8-4} = 2^4 = 16$ (1+1).	1+1+1+2
20	Number $= 3 \times 5^3$ (2); value 375 (2); it is $5^3 = 125$ times the original (1).	2+2+1
	OR: A is heavier (1); ratio $= \frac{1.8}{6} \times 10^{15-12} = 300$ times (2); total 1.806×10^{15} kg (2).	1+2+2

Mistakes to avoid:

- Writing $2^{-3} = -8$ or $-\frac{1}{8}$; a negative exponent means the **reciprocal**, so $2^{-3} = \frac{1}{8}$.
- Adding the bases: $2^3 \times 2^4 = 4^7$ instead of $2^{3+4} = 2^7$.
- Saying $a^0 = 0$; for $a \neq 0$ it is 1.
- Multiplying exponents when multiplying powers of the same base ($a^m \times a^n = a^{mn}$) or adding them in $(a^m)^n$.
- Leaving the answer as 25×10^4 or 0.25×10^6; in standard form the number part must satisfy $1 \leq m < 10$.
- Getting the sign of the exponent wrong for small numbers: $0.0003 = 3 \times 10^{-4}$, not 3×10^4.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Power Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 5	$x = 4$, so 5^1
2	**(d)** 3	$\left(\frac{1}{4} + \frac{1}{12}\right)^{-1}$
3	**(a)** 310×10^5	Convert each to usual form and compare
4	**(a)** 75	$5^{x+2} = 5^x \times 5^2$
5	**(b)** 5^4	$5 \times 5^3 = 5^4$
6	**(b)** 4	$4x - 1 = 15$
7	**(a)** 6	$2^3 \times 5^7 = 10^3 \times 625$
8	**(a)** 9	Ones digits repeat in the cycle [9, 1]
9	**(a)** Both A and R are true and R is the correct explanation of A.	Moving the decimal point 4 places to the right gives 4.5, so the number is 4.5×10^{-4}, which is R applied.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	A follows from $(a^m)^n = a^{mn}$; R is a different true law, so it does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Ones digits of powers of 6 repeat as [6] (1); 99 leaves position 1 in the cycle, so the digit is 6 (1).	1+1
12	$49 = 7^2$ (1); $3x - 1 = 2$ gives $x = 1$ (1).	1+1
13	Convert to the same power or usual form: 68000 (1); 6.8×10^4 (1).	1+1
14	Ones digits of powers of 8 repeat as [8, 4, 2, 6] (1); 102 leaves position 2 in the cycle, so the digit is 4 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Thickness doubles at each fold: 0.1×2^{15} (1); $= 3276.8$ mm (1); in cm $= 327.68$ cm (1).	1+1+1
16	15 min $= 900$ s (1); distance $= 3 \times 10^8 \times 900$ (1); $=2.7 \times 10^{11}$ m (1).	1+1+1
17	Take 7^2 common in the numerator: $7^2(7 + 1)$ (1); cancel $(7 + 1)$ (1); $7^2 \div 7^1 = 7$ (1).	1+1+1
	OR: Thickness doubles at each fold: 0.25×2^9 (1); $= 128$ mm (1); in cm $= 12.8$ cm (1).	1+1+1
18	20 min $= 1200$ s (1); distance $= 3 \times 10^8 \times 1200$ (1); $=3.6 \times 10^{11}$ m (1).	1+1+1
	OR: Take 6^3 common in the numerator: $6^3(6 + 1)$ (1); cancel $(6 + 1)$ (1); $6^3 \div 6^2 = 6$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{1}{2^4} = \frac{1}{16}$ (1). (ii) $\left(\frac{2}{3}\right)^2 = \frac{4}{9}$ (1). (iii) $7^{5-3} = 7^2 = 49$ (1). (iv) $2^{3-5-(-2)} = 2^0 = 1$ (1); laws $a^m \times a^n = a^{m+n}$ and $\frac{a^m}{a^n} = a^{m-n}$ (1).	1+1+1+2
20	Number $= 6 \times 2^7$ (2); value 768 (2); it is $2^7 = 128$ times the original (1).	2+2+1
	OR: A is heavier (1); ratio $= \frac{4.8}{1.2} \times 10^{18-15} = 4000$ times (2); total 4.8012×10^{18} kg (2).	1+2+2

Mistakes to avoid:

- Writing $2^{-3} = -8$ or $-\frac{1}{8}$; a negative exponent means the **reciprocal**, so $2^{-3} = \frac{1}{8}$.
- Adding the bases: $2^3 \times 2^4 = 4^7$ instead of $2^{3+4} = 2^7$.
- Saying $a^0 = 0$; for $a \neq 0$ it is 1.
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